

FUKUI, Japan

WMO: 476160

Lat: 36.0554N

Lon: 136.2230E

Elev: 57

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.0	(c) 30.4	(d) 20.4	(e) 15.3	(f) 40.5	(g) 22.8	(h) 17.2	(i) 39.6	(j) 18.9	(k) 38.5	(l) 17.0	(m) 38.8	(n) 5.7	(o) 190	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	93.6	76.9	91.7	76.5	89.6	76.0	78.9	88.6	78.0	87.2	77.1	86.0	10.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	138.5	82.9	75.5	133.8	82.2	74.6	129.8	81.8	42.5	88.4	41.5	87.5	40.7	86.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.5	(b) 15.7	(c) 13.7	(e) 25.9	(f) 96.9	(g) 1.8	(h) 1.4	(i) 24.6	(j) 97.9	(k) 23.5	(l) 98.7	(m) 22.5	(n) 99.6	(o) 21.2	(p) 100.6		
(a) 18.5	(b) 15.7	(c) 13.7	(e) 24.7	(f) 80.6	(g) 1.7	(h) 1.2	(i) 23.5	(j) 81.5	(k) 22.5	(l) 82.1	(m) 21.6	(n) 82.8	(o) 20.4	(p) 83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	38.0	39.0	45.2	55.4	65.0	72.0	79.8	82.3	74.3	63.7	52.7	43.0		
	DBStd	16.05	3.92	4.75	5.97	6.57	5.20	4.37	4.59	3.95	5.35	5.26	5.65	5.27		
	HDD50	1141	371	311	175	23	0	0	0	0	0	0	35	226		
	HDD65	3679	836	729	613	295	64	2	0	0	2	86	369	683		
	CDD50	4545	0	2	26	184	466	661	924	1000	730	426	117	9		
	CDD65	1607	0	0	0	6	65	213	459	535	281	47	1	0		
	CDH74	14116	0	0	0	58	432	1198	4406	5861	1971	188	2	0		
	CDH80	5257	0	0	0	7	69	284	1748	2521	605	23	0	0		
	WSAvg	6.2	5.8	6.1	6.7	7.0	6.9	5.8	5.9	6.2	6.0	5.7	5.8	5.9		
	PrecAvg	88.7	11.0	6.5	6.1	5.2	5.6	6.1	9.0	5.7	8.0	5.9	8.0	11.7		
(15) Precipitation	PrecMax	106.9	18.3	10.2	9.3	8.5	14.8	11.3	23.1	14.1	13.5	9.8	16.0	26.2		
	PrecMin	60.5	3.8	1.9	2.8	1.8	2.2	1.6	2.1	0.9	2.6	3.1	4.6	4.6		
	PrecStd	10.8	3.7	2.0	1.8	1.8	3.0	2.3	5.1	3.1	3.8	1.7	2.2	4.9		
	PrecStd	10.8	3.7	2.0	1.8	1.8	3.0	2.3	5.1	3.1	3.8	1.7	2.2	4.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	60.8	69.8	80.1	85.3	89.1	95.4	96.2	92.4	82.8	72.6	63.2		
		MCWB	46.6	50.8	54.0	60.0	65.4	72.5	77.3	77.5	75.4	70.1	60.2	54.3		
	2%	DB	50.0	54.9	64.6	74.6	81.8	86.0	93.3	94.3	89.4	78.7	68.6	58.1		
		MCWB	44.2	47.0	51.5	58.7	63.5	71.7	76.7	77.1	75.0	66.9	58.8	51.1		
	5%	DB	47.3	51.0	60.2	71.0	78.8	83.3	91.2	92.5	86.4	75.6	65.2	54.6		
		MCWB	42.5	44.4	49.9	57.2	62.3	69.9	76.4	77.0	74.0	65.1	57.3	49.5		
	10%	DB	44.7	47.6	56.4	67.4	75.8	80.7	88.7	90.6	83.4	72.8	62.4	51.6		
		MCWB	40.8	42.4	48.8	55.4	61.6	68.9	75.7	76.7	72.6	63.7	56.3	47.0		
	0.4%	WB	48.7	53.7	57.3	66.3	70.5	76.6	79.8	80.7	78.1	72.3	63.8	56.9		
		MCDB	52.1	59.0	63.6	72.4	76.9	84.4	90.4	90.8	87.2	79.0	67.8	60.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.3	48.5	54.1	61.5	67.6	74.7	78.3	79.4	76.5	69.6	61.2	52.8		
		MCDB	48.6	53.2	61.0	69.8	75.9	81.5	88.3	89.0	84.7	76.0	65.8	56.5		
	5%	WB	43.1	45.5	51.8	59.4	65.3	72.9	77.4	78.5	75.3	67.4	59.1	49.8		
		MCDB	46.4	49.7	58.2	67.3	73.6	79.5	87.0	87.8	83.3	72.5	63.7	54.0		
	10%	WB	41.1	42.7	49.2	57.2	63.7	71.1	76.5	77.8	74.1	65.4	57.0	47.5		
		MCDB	44.4	47.0	55.4	64.4	72.0	77.5	85.6	86.6	81.5	71.0	61.3	51.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	12.1	15.7	17.4	16.5	13.6	12.8	14.0	13.8	15.1	14.2	11.6		
		MCDBR	15.4	19.5	23.2	23.9	22.9	18.4	16.7	16.5	16.6	17.9	18.8	17.3		
		MCWBR	10.1	12.4	13.0	10.9	9.2	7.1	5.4	5.3	5.9	8.2	10.6	11.2		
		MCDBR	13.9	17.2	18.6	17.9	16.8	13.4	14.6	14.4	14.5	13.4	14.6	14.9		
	5% WB	MCWBR	10.2	12.3	12.4	10.0	8.6	6.1	5.3	5.3	5.8	7.2	9.6	10.8		
		MCWBR	10.2	12.3	12.4	10.0	8.6	6.1	5.3	5.3	5.8	7.2	9.6	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.435	0.464	0.528	0.550	0.551	0.573	0.567	0.539	0.486	0.452	0.434	0.435		
	taud		1.979	1.933	1.824	1.826	1.883	1.908	1.983	2.061	2.167	2.154	2.107	2.010		
	Ebn at Noon		226	238	234	238	240	234	238	245	240	227	217	217		
	Edh at Noon		44	51	63	66	63	62	57	51	44	41	38	40		
(44) All-Sky Solar Radiation	RadAvg		551	793	1161	1477	1656	1568	1547	1615	1256	1007	688	491		
	RadStd		45	75	96	126	178	131	226	170	105	84	58	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.81	+1.17	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page